

Excel Master Practice Plan — Questions, Hints & Answer Key

Use the companion Excel workbook for the 500-row dataset, lookup tables, cleaning data, practice questions, and formula answer examples.

1. Foundations & References

Level 1 - Navigation & References

Question: Convert Sales_Raw range to an Excel Table. Freeze the header row. Practice A1 and R1C1 reference styles.

Hint: Use View > Freeze Panes and Ctrl+T.

References

Question: In a new column calculate Gross Sales = Units*UnitPrice. Then calculate Net Sales using Discount. Use relative references correctly.

Hint: Gross = Units*UnitPrice; Net = Gross*(1-Discount).

Absolute / Mixed

Question: Put tax rate 18% in B1 and calculate tax for every order. Use an absolute reference for the tax rate.

Hint: Reference B1 as \$B\$1.

Named Ranges

Question: Create named ranges for Product_Master ProductID and UnitPrice. Use the names in a lookup formula.

Hint: Formulas > Name Manager.

Data Validation

Question: Create a dropdown for Region containing East, West, North, South. Create number validation for Rating 1-5.

Hint: Data > Data Validation.

2. Sorting & Filtering

Sorting

Question: Sort by Region, then Category, then Sales descending.

Hint: Use multi-level Sort.

Filtering

Question: Filter Sales > 5000 AND Region = East. Then filter Category = Mobile OR Laptop.

Hint: Use AutoFilter; OR can be handled with multiple criteria/Advanced Filter.

Advanced Filter

Question: Create a criteria range for Region=West AND Sales>5000 and copy results to another location.

Hint: Use Data > Advanced.

3. Text & Data Cleaning

Text

Question: Extract first 3 characters of ProductID; last 2 characters of State Code like TX-2345; extract product category from ProductName if applicable.

Hint: Use LEFT, RIGHT, MID, LEN.

Text Cleaning

Question: Clean CustomerName by removing extra spaces and standardizing case.

Hint: =PROPER(TRIM(F2))

Cleaning

Question: Remove non-printing characters and replace double spaces/selected characters where needed.

Hint: Use CLEAN, SUBSTITUTE.

4. Dates & Logical Functions

Dates

Question: Create Order Month, Order Year, Delivery Days, and Working Days from OrderDate to DeliveryDate.

Hint: DAY, MONTH, YEAR, NETWORKDAYS.

Date calculations

Question: Calculate employee tenure from HireDate to TODAY(), and month-end date for each order month.

Hint: DATEDIF and EOMONTH.

Logical

Question: Create Performance: Above Target / Below Target based on Sales vs Target.

Hint: =IF(Sales>Target,"Above Target","Below Target")

Logical

Question: Create a 5-grade rating based on Rating: 5=A, 4=B, 3=C, 2=D, 1=E. Use nested IF and/or IFS if available.

Hint: Nested IF.

AND/OR/XOR

Question: Flag orders that are (Sales>5000 AND Discount<=10%) OR Category=Mobile. Create a separate XOR example for two boolean conditions.

Hint: Use AND, OR, XOR.

IFERROR

Question: Create a safe lookup that returns "Not Found" instead of an error.

Hint: Wrap lookup in IFERROR.

5. Lookup Functions

VLOOKUP exact

Question: Using Product_Master, return ProductName and UnitPrice from ProductID.

Hint: =VLOOKUP(H2,Product_Master!\$A\$2:\$F\$13,2,FALSE)

VLOOKUP approximate

Question: Create a small commission-rate table by sales amount and use approximate VLOOKUP to assign a rate.

Hint: Sort thresholds ascending; use TRUE.

HLOOKUP

Question: Make a horizontal lookup table of Region targets and retrieve the target for a selected Region.

Hint: Use HLOOKUP exact match.

XLOOKUP

Question: Return ProductName, Brand, and Supplier from ProductID. Test a missing ProductID with a custom not-found message.

Hint: =XLOOKUP(H2,Product_Master!A:A,Product_Master!B:B,"Not Found")

Reverse XLOOKUP

Question: For a selected CustomerName, return the last matching OrderID. Use search mode -1.

Hint: XLOOKUP(...,-1)

INDEX MATCH

Question: Return UnitPrice for a selected ProductID using INDEX + MATCH.

Hint: =INDEX(Product_Master!E:E,MATCH(A2,Product_Master!A:A,0))

Two-way INDEX MATCH

Question: Build a small Region x Month sales summary and return the intersection for West + March.

Hint: INDEX with two MATCH functions.

6. Conditional Formatting

Conditional Formatting

Question: Highlight duplicate OrderIDs, Sales above Target, top 10 Sales, and low Ratings <=2.

Hint: Use Duplicate Values, formula rules, Top/Bottom Rules, icon sets.

7. Pivot Tables & PivotCharts

Pivot Basics

Question: Create PivotTables: Sales by Region; Sales by Category; Average Sales by Salesperson; Count of Orders by PaymentMethod.

Hint: Rows/Columns/Values.

Pivot Advanced

Question: Group OrderDate by Month and Quarter. Add slicers for Region and Category. Drill into a high-sales value.

Hint: PivotTable Analyze > Group/Slicer.

Pivot Calculated Field

Question: Create a calculated field for gross margin proxy = Sales - Units*Cost. Add a Cost column from a suitable assumption or Product_Master extension.

Hint: Use Calculated Field or helper column.

PivotCharts

Question: Create Bar, Line, Area, and Pie charts from PivotTables. Add slicers and test interactions.

Hint: PivotChart + slicers.

Data Cleaning

Question: Use Remove Duplicates, Text to Columns, Flash Fill, Find/Replace wildcards, and error checking on Cleaning_Practice.

Hint: Work on a copy so raw data stays intact.

8. Cleaning Tools & Power Query

Power Query

Question: Import Sales_Raw into Power Query. Change data types, trim/clean names, remove blank rows, and load the cleaned table.

Hint: Data > From Table/Range.

Power Query Merge

Question: Merge Sales_Raw with Product_Master using ProductID. Expand Brand, Supplier, and UnitPrice.

Hint: Merge Queries; Left Outer.

Power Query Append

Question: Create two small monthly sales tables and append them into one table. Resolve mismatched columns and nulls.

Hint: Append Queries.

9. Dashboard & Capstone

Dashboard KPIs

Question: Build KPI cards for Total Sales, Average Order Value, Max Order, Min Order, Order Count, and % Above Target.

Hint: Use formulas or Pivot measures.

Dashboard

Question: Create an executive dashboard with KPI cards, monthly sales trend, region sales, category sales, slicers for Region/Category, and a date timeline.

Hint: Use PivotTables + PivotCharts + Slicers + Timeline.

Final Case Study

Question: Start from Sales_Raw. Clean it, enrich it with Product_Master, create calculated columns, summary tables, PivotTables, PivotCharts, and a one-page dashboard. Write 5 factual insights from your results.

Hint: Do not change the raw data; keep a Cleaned_Data sheet.

Formula Answer Key

Task	Example answer
Gross Sales	=J2*K2
Net Sales	=J2*K2*(1-L2)
Tax with absolute reference	=M2*\$B\$1
Clean Name	=PROPER(TRIM(F2))
ProductID prefix	=LEFT(H2,4)
Code suffix	=RIGHT(U2,4) if U2 contains a code like TX-2345
Safe lookup	=IFERROR(XLOOKUP(H2,Product_Master!A:A,Product_Master!B:B),"Not Found")
Delivery Days	=P2-O2
Working Days	=NETWORKDAYS(O2,P2)
Order Year	=YEAR(B2)
Order Month	=MONTH(B2)
Month End	=EOMONTH(B2,0)
Performance	=IF(M2>N2,"Above Target","Below Target")
AND condition	=AND(M2>5000,L2<=10%)
OR condition	=OR(M2>5000,C2="East")
XOR condition	=XOR(C2="East",G2="Mobile")
VLOOKUP Product	=VLOOKUP(H2,Product_Master!\$A\$2:\$F\$13,2,FALSE)
VLOOKUP Price	=VLOOKUP(H2,Product_Master!\$A\$2:\$F\$13,5,FALSE)
XLOOKUP Price	=XLOOKUP(H2,Product_Master!A:A,Product_Master!E:E,"Not Found")
INDEX MATCH Price	=INDEX(Product_Master!E:E,MATCH(H2,Product_Master!A:A,0))
Pivot Sales by Region	Rows=Region; Values=Sales (Sum)
Pivot Average Salesperson	Rows=Salesperson; Values=Sales (Average)
Conditional Formatting	Use formula =\$M2>\$N2 to highlight Sales above Target across the row.

Important: PivotTable layouts and Power Query results depend on your workbook version and your exact transformations, so the answer key gives the expected setup/result rather than a single fixed cell-by-cell output.